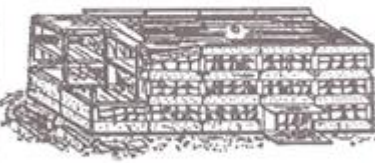
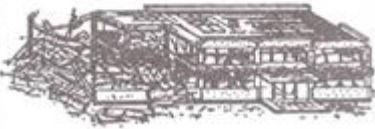


Source : Seismic retrofitting guidelines of buildings in Nepal (April 2013)

	Grade 3: Substantial to Heavy Damage (moderate structural damage, heavy non-structural damage) Cracks in columns and beam column joints of frames at the base and at joints of coupled walls. Spalling of concrete cover, buckling of reinforced rods. Large cracks in partition and infill walls, failure of individual infill panels.
	Grade 4: Very Heavy Damage (heavy structural damage, very heavy non structural damage) Large cracks in structural elements with compression failure of concrete and fracture of rebars; bond failure of beam reinforced bars; tilting of columns. Collapse of a few columns or of a single upper floor.
	Grade 5: Destruction (very heavy structural damage) Collapse of ground floor or parts (e. g. wings) of buildings.

S.No.	Damage Grade	Wall
1	Grade 1: Negligible to slight damage	No structural damage, slight non- structural damage - Hair line cracks in very few walls. - Fall of small pieces of plaster only. - Fall of loose stones from upper parts of buildings in very few cases
2	Grade 2: Moderate damage	Slight structural damage, moderate non-structural damage - Cracks in many walls. - Fall of fairly large pieces of plaster. - Partial collapse of chimneys.
3	Grade 3: Substantial to heavy damage	Moderate structural damage, heavy non- structural damage - Large and extensive cracks in most walls. - Roof tiles detach, chimney fracture at the roof line - Failure of individual non structural elements (partitions, gable walls, etc).
4	Grade 4: Very heavy damage	Heavy structural damage, very heavy non-structural damage - Serious failure of walls (gaps in walls) - Partial structural failure of roof and floors
5	Grade 5: Destruction	Very heavy structural damage